



英検2級 大問3B 説明文・伝記の長文読解 完全演習

全30セット / 問題編

THINKING 大問別問題集シリーズ | 本番そっくりの形式で30本の長文を演習。

- ・各長文（4～5段落）に内容一致・理由・指示語などの設問が5問（4択）。
 - ・**解き方のコツ**：まず設問(1)～(5)を先読み → 設問は本文の流れ順なので、段落を追いつながりキーワードに当たる箇所を探す → 選択肢は本文の**言い換え**。本文に書いていないことは選ばない。
 - ・解いたら巻末の解答で答え合わせ → **解説編**で「なぜ正解／なぜ誤りか」と解法を確認。
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B

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No.1 / Florence Nightingale (伝記：近代看護の母)

Florence Nightingale and Modern Nursing

Florence Nightingale is remembered today as the founder of modern nursing, but in her time, nursing was not seen as a respectable job. She was born into a wealthy British family who expected her to marry and live a quiet life at home. However, Nightingale felt that she had a duty to help others, and against her family's wishes, she decided to train as a nurse. This was a bold choice for a woman of her background.

Nightingale became famous during the Crimean War in the 1850s. She was asked to lead a group of nurses to care for British soldiers who had been wounded in battle. When she arrived at the military hospital, she was shocked by what she found. The building was dirty and overcrowded, and there was little clean water or fresh air. In fact, more soldiers were dying from diseases caught in the hospital than from their wounds.

Nightingale set to work at once. She and her nurses cleaned the wards, opened windows to let in fresh air, and made sure the men had clean clothing and better food. Within months, far fewer soldiers were dying. At night, she would walk through the dark hospital carrying a lamp to check on her patients, which is why she became known as "the Lady with the Lamp."

After the war, Nightingale did not stop working. She had collected detailed information about the soldiers' deaths, and she used charts and numbers to show officials that clean, well-run hospitals saved lives. Her reports helped change the way hospitals across the country were run. She also opened a training school for nurses in London, which helped turn nursing into a skilled and respected profession.

Although she spent much of her later life in poor health, Nightingale continued to write about how to improve health care. The ideas she fought for—cleanliness, fresh air, and careful record-keeping—are still important in hospitals today. Thanks to her work, nursing became a career that millions of people are proud to follow.

(1) Why was Nightingale's decision to become a nurse surprising?

- 1 Because she had no interest in helping others.
- 2 Because nursing was not seen as a respectable job for someone of her background.
- 3 Because she was too poor to train as a nurse.
- 4 Because her family wanted her to become a doctor.

(2) What shocked Nightingale at the military hospital?

- 1 The soldiers refused to be treated.
- 2 There were too few soldiers to care for.
- 3 The hospital was clean but very small.
- 4 Many soldiers were dying from diseases, not their wounds.

-
- (3)** Why was Nightingale called "the Lady with the Lamp"?
- 1** Because she went around the wards at night to see how her patients were.
 - 2** Because she invented a new kind of lamp.
 - 3** Because she only worked during the daytime.
 - 4** Because she taught soldiers to read by lamplight.
-

- (4)** How did Nightingale help change hospitals after the war?
- 1** She paid for new hospitals herself.
 - 2** She trained soldiers to become doctors.
 - 3** She used data to prove that well-run, clean hospitals saved lives.
 - 4** She closed down all the old hospitals.
-

- (5)** Which of the following is true?
- 1** Many of Nightingale's ideas are still used in hospitals today.
 - 2** Nightingale's family strongly supported her decision to become a nurse.
 - 3** Nightingale stopped all her work as soon as the war ended.
 - 4** Nightingale enjoyed perfect health throughout her later life.

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No.2 / The Return of the Wolves (科学: イエローストーンの狼)

The Return of the Wolves

Yellowstone National Park, in the United States, is home to many kinds of wildlife. About a hundred years ago, however, the park lost one of its most important animals: the gray wolf. People at the time saw wolves as dangerous pests that killed farm animals, so hunters were paid to get rid of them. By the 1920s, the wolves of Yellowstone had been completely wiped out.

At first, this seemed like good news for other animals, such as elk, a kind of large deer. With no wolves to hunt them, the elk population grew quickly. But this created a new problem. The huge number of elk ate almost all the young trees and plants, especially along the rivers. Without these plants, the land began to change, and many other animals that depended on them started to disappear.

For many years, scientists argued about whether wolves should be brought back. Finally, in 1995, a small number of wolves were captured in Canada and released into Yellowstone. No one was quite sure what would happen. The hope was that the wolves would reduce the elk population and help bring the park back into balance.

The results surprised even the scientists. As expected, the wolves hunted the elk, so their numbers fell. More importantly, the elk became more careful and avoided open areas near rivers where they could easily be caught. In these areas, young trees were finally able to grow tall again. Birds returned to the new trees, and beavers, which eat wood, came back too. The beavers built dams, which changed the way the rivers flowed.

The return of the wolves showed scientists something important: removing or adding a single animal can change a whole natural system in ways that are hard to predict. The wolves did not just affect the elk; they helped bring back plants, birds, and beavers, and even reshaped the rivers. Today, the story of Yellowstone's wolves is used around the world as an example of how closely living things are connected.

(1) Why were the wolves removed from Yellowstone?

- 1 Because they were taken to another park.
- 2 Because there was not enough food for them.
- 3 Because people believed they were a threat to their farm animals.
- 4 Because they had caught a disease.

(2) What problem did the growing elk population cause?

- 1 They destroyed much of the new plant life, especially near the rivers.
- 2 The elk attacked visitors to the park.
- 3 The elk moved away from the park.
- 4 The elk became too few in number.

(3) What did scientists hope the returning wolves would do?

- 1** Increase the number of elk.
 - 2** Make the park more dangerous.
 - 3** Bring more tourists to Yellowstone.
 - 4** Reduce the elk population and restore balance.
-

(4) According to the passage, what happened after the wolves returned?

- 1** The rivers dried up completely.
 - 2** Young trees were able to grow tall again.
 - 3** All the birds left the park.
 - 4** The elk began to spend more time near the rivers.
-

(5) Which of the following is true?

- 1** The wolves had no effect on any animals except the elk.
- 2** Scientists were certain about what would happen when the wolves returned.
- 3** The elk population grew even larger after the wolves returned.
- 4** Bringing back the wolves changed many parts of the park's natural system.

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No.3 / The Problem of Food Waste (社会問題：食品ロス)

The Problem of Food Waste

Around the world, an enormous amount of food is thrown away every year. In fact, experts believe that about a third of all the food produced for people is never eaten. This is a serious problem, not only because so much good food is lost, but also because of the harm that food waste causes to the planet.

Food is wasted at every stage, from the farm to the home. On farms, fruits and vegetables that are not perfectly shaped are often thrown away, even though they are perfectly safe to eat, because shops believe customers will not buy them. In shops, food that is close to its sell-by date may be removed from the shelves. The biggest waste of all, however, often happens in people's own homes, where families buy more than they need and let food go bad.

Wasting food does more harm than many people realize. Growing food uses huge amounts of water, energy, and land. When that food is thrown away, all of those resources are wasted too. What is worse, when food rots in large rubbish dumps, it produces gases that add to the warming of the Earth. At the same time, millions of people around the world do not have enough to eat.

The good news is that there are many ways to reduce food waste. Some shops now sell "ugly" fruits and vegetables at lower prices, and others give food that is close to its date to charities instead of throwing it away. At home, people can plan their meals, store food correctly, and use up leftovers. Even small changes, made by many people, can make a big difference.

One simple step is to understand date labels better. Many people throw food away as soon as the date on the package passes, but these dates often show only when food is at its best, not when it becomes unsafe. By learning to judge whether food is still good, and by buying only what they need, ordinary people can help cut food waste and protect both the planet and their own money.

(1) According to the passage, how much food produced for people is never eaten?

- 1 Roughly one in three portions of it.
- 2 About half of it.
- 3 Almost none of it.
- 4 All of it.

(2) Why are oddly shaped fruits and vegetables often thrown away on farms?

- 1 Because they are unsafe to eat.
- 2 Because there are too many of them to sell.
- 3 Because shops expect that shoppers will refuse to buy them.
- 4 Because they cost too much to grow.

(3) What is one way that food waste harms the planet?

- 1 It makes food more expensive to grow.
- 2 Food left to rot gives off gases that help heat up the Earth.
- 3 It uses up the world's supply of fruit.
- 4 It causes shops to close down.

-
- (4)** What is one thing some shops do to reduce food waste?
- 1** They stop selling fruit and vegetables.
 - 2** They throw away food more quickly.
 - 3** They buy less food from farms.
 - 4** They donate food that is near its date instead of discarding it.
-

- (5)** Which of the following is true?
- 1** Food is only wasted in people's homes.
 - 2** The dates printed on packages often mark a food's best quality rather than the point at which it stops being safe.
 - 3** There is nothing ordinary people can do about food waste.
 - 4** Growing food uses very few resources.

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No.4 / Louis Braille (伝記：点字の発明)

Louis Braille and the Gift of Reading

Louis Braille was born in France about two hundred years ago. As a small child, he loved to play in his father's workshop, where leather goods were made. One day, while playing with a sharp tool, he hurt one of his eyes in an accident. The injury became infected, and the infection soon spread to his other eye. By the age of about five, Louis had become completely blind.

Although he could no longer see, Louis was a bright and curious boy. He won a place at a special school for blind children in Paris. At that time, blind students were taught to read using books with large, raised printed letters that they felt with their fingers. However, this method was very slow and difficult, the books were huge and heavy, and blind people could not use it to write at all.

Louis was determined to find a better way. He heard about a system used by soldiers called "night writing," which used raised dots and dashes so that messages could be read in the dark without a light. The army's system was too complicated to be useful, but it gave Louis an important idea. He realized that a code made of raised dots might allow blind people to read quickly by touch.

Working patiently for several years, Louis created a simple system using just six raised dots. By arranging these dots in different patterns, he could represent every letter of the alphabet, as well as numbers and even musical notes. The system was easy to feel with a single fingertip, and, unlike the old method, blind people could write it themselves using a simple tool. Louis finished his system while he was still a teenager.

At first, Louis's invention was not widely accepted, and some teachers refused to use it. Sadly, it became truly popular only after his death. Today, however, the system known as "braille" is used by blind people all over the world. Thanks to a boy who refused to give up, millions of people who cannot see are able to read and write for themselves.

(1) How did Louis Braille become blind?

- 1 He was born without the ability to see.
- 2 He caught an illness at his school.
- 3 He was injured in a car accident.
- 4 He hurt his eye while playing with a tool.

(2) What was one problem with the reading method used at Louis's school?

- 1 The books were small and easy to lose.
- 2 It was slow and could not be used for writing.
- 3 Only teachers were allowed to use it.
- 4 It required electric light to read.

-
- (3)** What gave Louis the idea for his system?
- 1** A book written by another blind student.
 - 2** A new kind of printing machine.
 - 3** A "night writing" system used by soldiers.
 - 4** A suggestion from his teacher.
-

- (4)** What was special about the system Louis created?
- 1** Unlike the older method, it let blind people produce writing on their own.
 - 2** It used hundreds of different shapes.
 - 3** It could only be used to write music.
 - 4** It needed both hands to read.
-

- (5)** Which of the following is true?
- 1** Louis's invention became popular during his own lifetime.
 - 2** The army's night-writing system was simple and easy to use.
 - 3** Braille is now used by blind people around the world.
 - 4** Louis gave up trying to improve reading for the blind.

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No.5 / Why Cities Are Hotter (科学・社会：ヒートアイランド)

Why Cities Are Hotter

On a hot summer day, a large city can be several degrees warmer than the countryside around it. Scientists call this difference the "urban heat island" effect. Even at night, when the countryside cools down, the city often stays warm. For people living in cities, this can make hot weather feel even more uncomfortable and, in some cases, dangerous.

There are several reasons why cities become so hot. One of the most important is the materials that cities are built from. Roads and buildings are usually made of dark concrete, brick, and asphalt. These dark surfaces absorb a lot of heat from the sun during the day and then release it slowly through the evening and night. Unlike soil or grass, they do not allow the ground to cool easily.

Cities also have far fewer trees and plants than the countryside. Plants help cool the air, both by providing shade and by releasing water from their leaves. With less greenery, cities lose this natural cooling. On top of this, the many cars, factories, and air conditioners in a city all give off heat of their own, making the problem even worse.

The urban heat island effect causes real problems. During very hot weather, it can put people's health at risk, especially the elderly and the very young. It also leads people to use their air conditioners more, which uses more electricity. Burning extra fuel to make this electricity can create more pollution and release more of the gases that warm the planet—so the problem feeds itself.

Fortunately, cities can take steps to cool down. Planting more trees and creating parks adds shade and natural cooling. Some cities are painting roofs white or covering them with plants, known as "green roofs," so that buildings absorb less heat. Using lighter-colored materials for roads can also help. By making changes like these, cities can become more comfortable and healthier places to live, even as summers grow hotter.

(1) What is the "urban heat island" effect?

- 1 A city being warmer than the countryside around it.
- 2 A city being colder at night than during the day.
- 3 An island that is warmer than the sea.
- 4 A type of summer storm over cities.

(2) Why do dark roads and buildings make cities hotter?

- 1 They block the wind from cooling the city.
- 2 They are usually covered in grass.
- 3 They reflect sunlight back into space.
- 4 They absorb heat by day and release it slowly.

(3) How do plants help cool the air?

- 1** They produce electricity for the city.
 - 2** They give shade and release water from their leaves.
 - 3** They absorb heat from cars and factories.
 - 4** They make the city's roads darker.
-

(4) According to the passage, why does the heat island problem "feed itself"?

- 1** Because people plant more trees in hot weather.
 - 2** Because cities become cooler over time.
 - 3** Because using more air conditioning leads to more pollution and warming.
 - 4** Because people leave the city in summer.
-

(5) Which of the following is true?

- 1** Cities and the countryside are always exactly the same temperature.
- 2** Plants make cities hotter by blocking the wind.
- 3** Nothing can be done to cool down hot cities.
- 4** Painting roofs white can help buildings absorb less heat.

3

B

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No.6 / Wangari Maathai (伝記：植林とノーベル平和賞)

Wangari Maathai and the Green Belt Movement

Wangari Maathai grew up in a rural village in Kenya, surrounded by green forests, clean streams, and rich farmland. She was a clever student and was lucky enough to study science, first in Kenya and later abroad. When she returned home as an adult, however, she was saddened to see how much her country had changed. Many of the forests she remembered had been cut down.

Maathai understood that the loss of the trees was causing serious problems. Without forests, the soil was washed away by the rain, streams dried up, and it became harder to grow food or find clean water and firewood. She noticed that rural women suffered the most, because it was usually their job to find water and fuel for their families. She began to wonder what could be done.

Maathai's answer was simple but powerful: plant trees. In the 1970s, she started an organization called the Green Belt Movement, which encouraged ordinary women to plant trees in their villages. The women were paid a small amount for each tree that grew, which also gave them an income. Over the years, the movement planted tens of millions of trees across Kenya.

Maathai's work was not always welcomed. Some powerful people did not like a woman speaking out, and at times the government tried to stop her. She was even arrested for protesting against plans to cut down forests for building projects. But Maathai refused to give up, and her courage inspired many others to join her cause.

In 2004, Wangari Maathai was given the Nobel Peace Prize, becoming the first African woman to receive it. The prize recognized that caring for the environment, helping the poor, and building peace are all connected. Maathai once said that planting a tree is a simple act of hope. Long after her death, her movement continues, and her message—that ordinary people can heal the Earth—still inspires the world.

- (1) How did Maathai feel when she returned home as an adult?
- 1 She was proud of how her village had grown.
 - 2 She was excited to start a new job.
 - 3 She was upset by the damage to the natural areas she knew.
 - 4 She no longer remembered her village.

- (2) Why did rural women suffer the most from the loss of trees?
- 1 Collecting water and fuel was usually their responsibility.
 - 2 They were not allowed to plant trees.
 - 3 They had to move to the cities.
 - 4 They owned all the forests.

- (3) What did the Green Belt Movement encourage women to do?
- 1 Move away from their villages.
 - 2 Stop using firewood.
 - 3 Build new schools.
 - 4 Grow trees and make a little money by doing so.

(4) What happened because of Maathai's work?

- 1** The government gave her an award at once.
 - 2** She was taken into custody for opposing forest clearing.
 - 3** She stopped her work for many years.
 - 4** Everyone immediately agreed with her.
-

(5) Which of the following is true?

- 1** No African woman had won the Nobel Peace Prize before her.
- 2** The Green Belt Movement planted only a few trees.
- 3** Maathai gave up her work after she was arrested.
- 4** Maathai believed that protecting nature and peace were unrelated.

3

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No.7 / The Secret Life of Octopuses (科学：タコの知能)

The Secret Life of Octopuses

When we think of clever animals, we usually picture dogs, dolphins, or chimpanzees. Yet one of the most intelligent creatures in the ocean is one that looks nothing like us: the octopus. Although it is a soft-bodied sea animal with no bones at all, the octopus can solve problems in ways that surprise even scientists.

In experiments and aquariums, octopuses have shown remarkable skills. They can open jars to reach food inside, find their way through mazes, and even escape from closed tanks by squeezing through tiny gaps. Some octopuses have been seen carrying empty coconut shells across the sea floor, later using them as a kind of shelter—a clear example of using a tool, something once thought to be possible only for "higher" animals.

The octopus also has an amazing way of protecting itself. In less than a second, it can change the color and even the texture of its skin to match its surroundings. This lets it hide from enemies by blending into rocks, sand, or coral. If it is discovered, it can shoot out a cloud of dark ink to confuse the attacker and then escape.

Perhaps the strangest thing about an octopus is its body. An octopus has three hearts and blue blood. Even more surprising, much of its nervous system is not in its head but in its eight arms. Each arm contains many nerve cells and can touch, taste, and move almost on its own. In a sense, an octopus can "think" with its arms as well as its brain.

Despite all these abilities, the octopus remains something of a mystery. Most kinds live only a year or two, so they have little time to learn from older octopuses, and they mostly live alone. Scientists are still trying to understand how such a short-lived, solitary animal became so clever. The more we study the octopus, the more we realize how many different forms intelligence can take.

(1) What is surprising about the octopus?

- 1 It looks exactly like a dolphin.
- 2 It is very intelligent despite having no bones.
- 3 It is the largest animal in the ocean.
- 4 It cannot solve any problems.

(2) What is one example of an octopus using a tool?

- 1 Building a nest out of seaweed.
- 2 Catching fish with a net.
- 3 Writing messages in the sand.
- 4 Carrying coconut shells to use as shelter.

-
- (3)** How does an octopus avoid being seen by its enemies?
- 1** By making its body look like the rocks and sand around it.
 - 2** By swimming faster than any fish.
 - 3** By making a loud sound to scare them.
 - 4** By covering itself with a hard shell.
-
- (4)** According to the passage, what is unusual about an octopus's nervous system?
- 1** It is found only in the head.
 - 2** It does not exist at all.
 - 3** Much of it is in the octopus's arms.
 - 4** It stops working at night.
-
- (5)** Which of the following is true?
- 1** Octopuses usually live for many years.
 - 2** Octopuses learn most of their skills from older octopuses.
 - 3** Octopuses use a hard shell to protect themselves.
 - 4** Scientists are still trying to understand why octopuses are so clever.

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No.8 / 健康・社会：10代の睡眠

Why Teenagers Need More Sleep

Many adults believe that teenagers stay up late and sleep in simply because they are lazy. Scientists, however, have discovered that this is not the whole truth. As children grow into teenagers, a natural change takes place inside their bodies that makes them feel sleepy later at night and want to wake up later in the morning.

Every person has an internal "body clock" that controls when we feel awake and when we feel tired. During the teenage years, this clock shifts. A teenager's body often does not feel ready for sleep until quite late, sometimes around midnight. Because teenagers need about eight to ten hours of sleep to stay healthy, this means many of them would naturally wake up later in the morning than younger children or adults.

Here lies the problem. Most schools start early in the morning, long before a teenager's body clock says it is time to be awake. As a result, many teenagers do not get enough sleep on school nights. Going to school tired makes it harder to concentrate, to remember what they learn, and to control their emotions. Some studies even link a lack of sleep to lower grades and a greater chance of feeling unhappy.

Because of findings like these, some schools have decided to start their classes later in the morning. The results have been encouraging. At schools that pushed back their start times, students reported feeling more awake, and in some cases their grades and attendance improved. Teachers also noticed that students seemed more cheerful and ready to learn.

Changing the time school starts is not always easy, as it affects buses, parents' work, and after-school activities. Still, scientists agree that helping teenagers get enough sleep is important for their health and learning. Until school times change, experts suggest that teenagers can help themselves by keeping regular sleeping hours and avoiding bright screens late at night, which can trick the brain into staying awake.

(1) What have scientists discovered about teenagers and sleep?

- 1 Teenagers do not need much sleep.
- 2 Teenagers are simply lazy.
- 3 Teenagers should go to bed earlier than children.
- 4 A natural body change makes them feel sleepy later.

(2) How much sleep do teenagers need to stay healthy?

- 1 About four to five hours.
- 2 About eight to ten hours.
- 3 As little as possible.
- 4 The same as young children.

(3) Why do many teenagers not get enough sleep on school nights?

- 1** Because they sleep too much during the day.
 - 2** Because they are not interested in school.
 - 3** Because schools start before their body clock is ready.
 - 4** Because they have no homework.
-

(4) What happened at schools that started classes later?

- 1** Students felt more awake, and grades sometimes improved.
 - 2** Students became more tired than before.
 - 3** The schools soon closed down.
 - 4** Teachers asked to start even earlier.
-

(5) Which of the following is true?

- 1** Scientists believe teenagers are just being lazy.
- 2** Bright screens late at night can make it harder to fall asleep.
- 3** Changing school start times is always quick and simple.
- 4** Lack of sleep has no effect on teenagers' grades.

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No.9 / The Wright Brothers (伝記：動力飛行の発明)

The Brothers Who Taught the World to Fly

For thousands of years, people dreamed of flying like birds, but no one had built a machine that could carry a person through the air under its own power. At the start of the twentieth century, two American brothers, Wilbur and Orville Wright, set out to change that. They were not famous scientists; they ran a small shop that built and repaired bicycles.

The brothers approached the problem carefully. They read everything they could find about flight, and they spent hours watching how birds turned and balanced in the air. They realized that the hardest part of flying was not lifting a machine off the ground, but controlling it once it was in the air. Many earlier inventors had ignored this problem, but the Wrights made it the center of their work.

To test their ideas, the brothers first built kites and gliders, flying them again and again to see what worked. When they were unsure which wing shapes were best, they built their own small wind tunnel—a box that blew air over model wings—so they could test many designs quickly and safely. Step by step, through careful experiments, they slowly solved the problems of flight.

Finally, in 1903, on a windy beach at Kitty Hawk, North Carolina, the brothers were ready. Their aircraft, powered by a small engine they had built themselves, lifted into the air with Orville lying at the controls. The flight lasted only twelve seconds and covered a short distance, but it was the first time a powered machine had carried a person in controlled flight. History had been made.

At first, few people believed what the brothers had done, and the news spread slowly. But the Wrights kept improving their machines until they could fly for much longer. Within a few years, the whole world understood that human flight was truly possible. The careful, patient way the brothers worked—testing each idea before moving on—shows how a difficult dream can be reached one small step at a time.

(1) What were the Wright brothers' jobs before they worked on flight?

- 1 They owned a small business that made and fixed bicycles.
- 2 They were famous university scientists.
- 3 They were airplane pilots.
- 4 They worked as birdwatchers.

(2) What did the brothers realize was the hardest part of flying?

- 1 Making a machine that was large enough.
- 2 Finding a place with enough wind.
- 3 Keeping the machine steady after it was already flying.
- 4 Learning what birds usually ate.

(3) Why did the brothers build their own wind tunnel?

- 1 To keep warm during the winter.
- 2 To test many wing designs quickly.
- 3 To fly to other cities.
- 4 To teach other people about flight.

(4) What was special about the flight in 1903?

- 1** It lasted for several hours.
 - 2** It was flown without any engine.
 - 3** It carried many passengers.
 - 4** It was the first powered, controlled flight carrying a person.
-

(5) Which of the following is true?

- 1** The brothers gave up after their first short flight.
- 2** Many people believed the news immediately.
- 3** The brothers solved the problems of flight through careful experiments.
- 4** The Wrights were already famous scientists.

3

B

次の英文の内容に関して、(1) から (5) までの質問に対して最も適切なもの、または文を完成させるのに最も適切なものを 1, 2, 3, 4 の中から一つ選びなさい。

No.10 / The Trouble with Ocean Plastic (社会・環境：海洋プラスチック)

The Trouble with Ocean Plastic

Plastic is cheap, light, and useful, which is why we use it for everything from bottles to bags. Unfortunately, a huge amount of this plastic ends up in the ocean. Every year, millions of tons of plastic waste are washed or thrown into the sea, where it can travel thousands of kilometers on the ocean's currents. Some of it gathers in enormous floating patches far from any land.

One of the biggest problems with plastic is that it does not rot away like natural materials. Instead, sunlight and waves slowly break it into smaller and smaller pieces. The tiniest of these pieces, called microplastics, are smaller than a grain of rice. Because they are so small, they are almost impossible to clean up, and they can now be found in seawater all over the world, even in the deepest parts of the ocean.

Ocean plastic is very dangerous for sea life. Larger pieces can trap animals such as turtles, seals, and seabirds, making it hard for them to swim or breathe. Many animals also mistake plastic for food and eat it. A sea turtle, for example, may swallow a floating plastic bag, thinking it is a jellyfish. Plastic in their stomachs can make animals ill or even kill them.

The problem does not stop with sea animals. When small fish eat microplastics, and bigger fish eat those small fish, the plastic moves up the food chain. Some of these fish are later caught and eaten by people. Scientists are still studying what effect this may have on human health, but many believe that the plastic we throw away may eventually return to us on our own plates.

The good news is that there are ways to help. The most important is to use less plastic, especially "single-use" items like straws and bags that are thrown away after one use. People can also join beach clean-ups, and countries can build better systems to collect and recycle waste before it reaches the sea. The ocean's problems took many years to create, but with effort, we can begin to solve them.

(1) What happens to much of the plastic that reaches the ocean?

- 1 It sinks immediately to the bottom.
- 2 It travels far on the ocean's currents.
- 3 It turns back into oil.
- 4 It is quickly cleaned up by ships.

(2) What are "microplastics"?

- 1 A type of sea animal.
- 2 Plastic that rots away quickly.
- 3 Large floating islands of plastic.
- 4 Tiny pieces of broken-down plastic.

-
- (3)** Why might a sea turtle eat a plastic bag?
- 1** It thinks the bag is something it can eat.
 - 2** It enjoys the taste of plastic.
 - 3** It uses the bag to build a nest.
 - 4** It is trying to clean the ocean.
-
- (4)** How can ocean plastic reach people?
- 1** By floating onto beaches where people walk.
 - 2** By turning into clean water.
 - 3** By moving up the food chain into fish that people eat.
 - 4** By being used to make new bottles.
-
- (5)** Which of the following is true?
- 1** Plastic in the ocean rots away within a few days.
 - 2** Microplastics are easy to clean up.
 - 3** Ocean plastic is harmful only to large animals.
 - 4** Using fewer single-use items can help reduce ocean plastic.

3

B

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No.11 / Living on the Space Station (科学・宇宙：ISSの暮らし)

Living on the Space Station

High above the Earth, a spacecraft about the size of a football field circles the planet again and again. This is the International Space Station, or ISS, where astronauts from many countries live and work for months at a time. Travelling at enormous speed, the station orbits the Earth roughly once every ninety minutes, so the astronauts on board see the sun rise and set about sixteen times a day.

The most unusual thing about life on the ISS is that everything floats. Because the station is falling around the Earth, the astronauts and their belongings feel almost no weight, a condition often called "weightlessness." This makes simple tasks strange and new. Astronauts cannot pour a drink into a cup, for example, so they drink from sealed bags through a straw. They must also tie down their tools, or the tools will simply drift away.

Eating and sleeping are different too. Much of the food is dried or specially packaged so that it does not make a mess, and astronauts add water to many meals before eating them. At bedtime, there are no beds to lie on. Instead, astronauts climb into sleeping bags that are attached to the walls, so that they do not float around the cabin while they sleep.

Living without gravity may sound fun, but it is hard on the body. On Earth, our muscles and bones stay strong because they work against gravity all day. In space, they are barely used, so they grow weaker over time. To stay healthy, astronauts must exercise for about two hours every day, using special machines that hold them in place while they run or lift weights.

The station must also be careful with resources. Water is so valuable that almost all of it is recycled and used again, even the water in the air the astronauts breathe out. Living on the ISS is not easy, but it allows scientists to do experiments that are impossible on Earth and helps us learn what humans will need for longer journeys, such as a future trip to Mars.

(1) How often does the ISS orbit the Earth?

- 1 About once every ninety minutes.
- 2 About once a day.
- 3 About once a week.
- 4 About once a year.

(2) Why do astronauts drink from sealed bags through a straw?

- 1 Because there is no water on the station.
- 2 Because the drinks are too hot.
- 3 Because liquids float and cannot be poured into a cup.
- 4 Because they are not allowed to use cups on Earth.

(3) How do astronauts keep from floating away while they sleep?

- 1** They lie on heavy beds fixed to the floor.
 - 2** They get into sleeping bags fastened to the walls.
 - 3** They hold on to their tools all night.
 - 4** They take turns staying awake to watch each other.
-

(4) Why must astronauts exercise for about two hours a day?

- 1** To pass the time on the station.
 - 2** To keep the station clean.
 - 3** To make the food taste better.
 - 4** Because their muscles and bones grow weaker in space.
-

(5) What is one thing the passage says about the ISS?

- 1** Nearly all of its water is cleaned and used a second time.
- 2** Astronauts can easily pour drinks into cups.
- 3** Living in space makes the human body stronger.
- 4** It teaches us nothing useful about future space travel.

3

B

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No.12 / Hokusai (伝記：葛飾北斎・生涯学び続けた絵師)

Hokusai: The Artist Who Never Stopped Learning

About two hundred years ago, in the city of Edo—today's Tokyo—lived one of Japan's greatest artists, Katsushika Hokusai. He worked during a time when colorful woodblock prints, called ukiyo-e, were popular and affordable, allowing ordinary people, not just the rich, to enjoy art. Over his long life, Hokusai created an enormous number of works—paintings, drawings, and prints—numbering in the thousands.

Hokusai's most famous work is a print known around the world as "The Great Wave off Kanagawa." It shows a towering wave about to crash down on small boats, with the distant Mount Fuji seen between the wave and the sky. This print was part of a series called Thirty-six Views of Mount Fuji, in which Hokusai drew the famous mountain from many different places and in many kinds of weather.

What made Hokusai unusual was that he never felt his work was good enough. Even as an old man, he kept studying and trying to improve. He once wrote that nothing he had made before the age of seventy was really worth much, and he believed that if he could live longer, his art would become better and better. This love of learning drove him to keep working almost until the end of his life.

Many years after Hokusai's death, his prints traveled to Europe, where they amazed Western artists. At that time, few Europeans had seen Japanese art, and Hokusai's bold designs and use of color were unlike anything they knew. Famous painters studied his work and borrowed ideas from it, and his influence can be seen in the art of that period.

Today, "The Great Wave" is one of the most recognized images in the world, printed on everything from books to cups. Hokusai is remembered not only for his beautiful art but also for his attitude: the belief that a person can always keep growing, no matter how old they are. His life is a reminder that it is never too late to improve at something we love.

(1) During Hokusai's time, why could ordinary people enjoy art?

- 1 Because all art was given away for free.
- 2 Because only the rich were allowed to buy it.
- 3 Because cheap printed pictures put art within reach of common people.
- 4 Because there were no other forms of entertainment.

(2) What does "The Great Wave off Kanagawa" show?

- 1 A huge wave above small boats, with Mount Fuji behind.
- 2 A quiet lake on a sunny day.
- 3 A busy city street in Edo.
- 4 A garden full of flowers.

(3) What was unusual about Hokusai's attitude to his work?

- 1** He thought his early work was perfect.
 - 2** He believed he could always keep improving.
 - 3** He stopped drawing when he was young.
 - 4** He never showed his work to anyone.
-

(4) What happened when Hokusai's prints reached Europe?

- 1** They were ignored by everyone.
 - 2** They were quickly forgotten.
 - 3** They amazed Western artists, who borrowed ideas from them.
 - 4** They were used only to wrap goods.
-

(5) Which of the following is true?

- 1** Hokusai believed people can keep growing no matter how old they are.
- 2** Hokusai created only a few works in his life.
- 3** Hokusai's art had no effect on European painters.
- 4** "The Great Wave" is little known today.

3

B

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No.13 / The Monarch Butterfly (科学・自然：オオカバマダラの移動)

The Amazing Journey of the Monarch Butterfly

Each autumn, in eastern North America, millions of orange-and-black monarch butterflies begin one of the most amazing journeys in nature. As the weather turns cold, they fly south, travelling thousands of kilometers to spend the winter in the warm forests of central Mexico. For such small and delicate creatures, this is an incredible feat of endurance.

The strangest part of the story is that no single butterfly makes the whole round trip. The monarchs that fly south in autumn live for several months, but the ones that travel back north in spring lay eggs and die along the way. It takes several generations of butterflies—children, grandchildren, and great-grandchildren—to complete the journey north again. In other words, the butterflies that return to Mexico the next winter have never been there before.

This raises a puzzling question: how do the butterflies know where to go if they have never made the journey? Scientists are not completely sure, but they believe the monarchs use the position of the sun, and perhaps the Earth's magnetic field, to find their direction. Somehow, the route to the same Mexican forests is passed down through the generations, even though no butterfly can be taught by its parents.

Sadly, monarch numbers have fallen in recent years. One major reason is the loss of a plant called milkweed. Monarch caterpillars can eat the leaves of only this one plant, so without it, they cannot grow. As farms and cities have spread, much of the milkweed has disappeared. Changes in weather and the loss of their winter forests have also made life harder for the butterflies.

The good news is that people are trying to help. Many families and schools now plant milkweed and flowers in their gardens to give the monarchs food along their route. Protecting the forests in Mexico is also important. If enough people take action, this tiny traveler may continue its remarkable journey for generations to come.

(1) Where do the monarch butterflies spend the winter?

- 1 In cold areas far to the north.
- 2 In warm woodlands in central Mexico.
- 3 High up in cold mountains.
- 4 In the place where they hatched.

(2) What is the strangest part of the monarchs' journey?

- 1 Not one butterfly travels the full route there and back.
- 2 The butterflies move only after dark.
- 3 The butterflies stay in Mexico all year.
- 4 The whole trip is over in just a few days.

(3) How do scientists think the monarchs find their way?

- 1** They follow other kinds of animals.
 - 2** They recall the trip they made before.
 - 3** They rely on the sun and possibly the Earth's magnetism.
 - 4** They learn the route from their parents.
-

(4) Why is the loss of milkweed a problem for monarchs?

- 1** Their young feed on no other plant.
 - 2** The adults sleep inside its flowers.
 - 3** It keeps them safe from the cold.
 - 4** It allows them to fly more quickly.
-

(5) Which of the following is true?

- 1** A single butterfly completes the whole journey by itself.
- 2** Monarch numbers have grown quickly in recent years.
- 3** Growing milkweed and flowers can support the monarchs.
- 4** Scientists are completely certain how monarchs navigate.

3

B

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No.14 / Eliud Kipchoge (伝記・スポーツ：2時間切りとマインドセット)

Eliud Kipchoge and the Two-Hour Marathon

A marathon is one of the hardest races in sport, in which runners cover a distance of just over forty-two kilometers. For many years, people wondered whether a human could ever run it in under two hours. The man who came closest is Eliud Kipchoge, a long-distance runner from Kenya who is widely seen as one of the greatest marathon runners in history.

Kipchoge has won many of the world's biggest races, including Olympic gold medals. In 2019, in a specially planned event in Vienna, he finally did what many had thought impossible: he ran the marathon distance in a time of one hour, fifty-nine minutes, and forty seconds—under two hours. Because the event used special helpers to set the pace and was not a normal race, the time was not counted as an official world record. Even so, it proved that the two-hour barrier could be broken.

What makes Kipchoge special is not only his strong body but also his calm and disciplined mind. He is famous for saying, "No human is limited," meaning that people can achieve far more than they think if they truly believe in themselves and work hard. He trains carefully, lives simply, and stays patient, even when a race becomes painful.

Kipchoge follows a strict daily routine. He wakes early to train, sometimes running well over a hundred kilometers in a single week, and he rests and eats carefully to let his body recover. He even shares simple jobs, like cleaning, with the other runners at his training camp, believing that no one should think they are too important to do ordinary work.

Kipchoge's achievements have inspired runners and non-runners alike. His story shows that great success comes not from talent alone, but from years of patient effort, self-control, and belief. Whether or not anyone soon runs an official marathon under two hours, Kipchoge has already changed what people believe is possible.

(1) What did people wonder about the marathon for many years?

- 1 Whether the distance could be made shorter.
- 2 Whether anyone could finish it in less than two hours.
- 3 Whether it was too easy a race.
- 4 Whether anyone would want to run it.

(2) Why was Kipchoge's under-two-hour time not counted as an official world record?

- 1 Because he ran the wrong distance.
- 2 Because he was too slow.
- 3 Because no one was watching.
- 4 Because the run was specially arranged rather than a regular race.

-
- (3)** What does Kipchoge's saying "No human is limited" mean?
- 1** People can do more than they imagine if they believe and try hard.
 - 2** Only a few people are born to win.
 - 3** Humans should never try difficult things.
 - 4** Running is bad for the body.
-
- (4)** What does Kipchoge do at his training camp?
- 1** He refuses to do any ordinary jobs.
 - 2** He shares simple jobs like cleaning with other runners.
 - 3** He trains only once a week.
 - 4** He eats whatever he likes.
-
- (5)** Which of the following is true?
- 1** Kipchoge believes success comes from talent alone.
 - 2** Kipchoge has never won a major race.
 - 3** Kipchoge showed that running the distance in under two hours was possible.
 - 4** Kipchoge trains only when he feels like it.

3

B

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No.15 / How the Bicycle Changed the World (社会史：自転車が社会を変えた)

How the Bicycle Changed the World

Today we think of the bicycle as a simple, everyday machine, used for exercise or short trips. When it first became popular in the late nineteenth century, however, the bicycle was something remarkable. For the first time, ordinary people had a cheap and easy way to travel longer distances on their own, without walking and without needing a horse.

Before the bicycle, most people who wanted to travel had to walk or, if they were wealthy, keep a horse. A horse was expensive: it had to be fed, housed, and cared for every day. A bicycle, by contrast, needed none of this. Once a person owned one, they could ride wherever they liked, whenever they wished, at almost no cost. People could suddenly visit nearby towns, reach jobs further from home, and meet people they would never have met before.

The bicycle had an especially large effect on the lives of women. At that time, women were expected to stay close to home and were not free to travel alone. The bicycle gave them a new independence: a young woman could now ride out by herself, without needing anyone's permission or company. The bicycle even changed the way women dressed, as the long, heavy skirts of the time were not practical for riding, and simpler, more comfortable clothes became common.

The popularity of the bicycle brought other changes too. Because so many people wanted smooth surfaces to ride on, there was new pressure to build better roads. The skills used to build bicycles—working with metal tubes, gears, and chains—later helped engineers build other machines, including early cars and even airplanes. In this way, the humble bicycle helped pave the way for the transport of the future.

It is easy to forget how important the bicycle once was. Yet for millions of people, it was the first taste of true freedom to move about as they pleased. Even today, in many parts of the world, the bicycle remains one of the cheapest, healthiest, and most useful ways to travel—proof that a simple idea can change society in lasting ways.

(1) Why was the bicycle remarkable when it first became popular?

- 1 It let ordinary people travel longer distances cheaply and on their own.
- 2 It was faster than any car.
- 3 It could only be used by the wealthy.
- 4 It was used mainly for sport.

(2) Why was keeping a horse difficult for most people?

- 1 Horses were not allowed in towns.
- 2 Looking after one every day cost a great deal of money.
- 3 Horses could not travel long distances.
- 4 Horses were faster than bicycles.

(3) How did the bicycle affect women's lives?

- 1** It made them stay closer to home.
 - 2** It forced them to wear heavier clothes.
 - 3** It gave them new independence to travel alone.
 - 4** It had no effect on women.
-

(4) According to the passage, what was one wider effect of the bicycle?

- 1** Roads were no longer needed.
 - 2** People stopped building machines.
 - 3** Horses became more popular than ever.
 - 4** The know-how gained from making bicycles was later used to build cars and aircraft.
-

(5) Which of the following is true?

- 1** The bicycle was useful only to wealthy people.
- 2** The bicycle gave many people a new freedom to travel.
- 3** The bicycle made it harder for women to go out.
- 4** The bicycle is no longer used anywhere today.

3

B

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No.16 / The Discovery of Penicillin (医学・歴史：ペニシリンの偶然の発見)

A Lucky Accident: The Discovery of Penicillin

Some of the greatest discoveries in science have happened by accident. One of the most famous examples is the discovery of penicillin, a medicine that has saved millions of lives. The story begins with a Scottish scientist named Alexander Fleming, who studied the tiny living things called bacteria that can cause illness. Fleming was known as a brilliant but rather untidy worker.

One day in 1928, Fleming returned to his laboratory after being away and noticed something strange. He had left a dish of bacteria uncovered, and a kind of mold had begun to grow on it. Most scientists might simply have thrown the dish away. But Fleming looked closely and saw that, in the area around the mold, the bacteria had died. Something produced by the mold seemed to be killing them.

Fleming realized that this was important. He carried out tests and found that the mold made a substance that could kill many kinds of harmful bacteria. He named this substance penicillin. He understood that it might one day be used as a medicine, and he wrote about his discovery so that other scientists could read it. However, Fleming was not able to produce penicillin in large enough amounts to treat patients.

For about ten years, little was done with Fleming's discovery. Then, a team of other scientists found ways to make penicillin in much larger amounts and proved that it could cure dangerous infections in people. During and after a great war, penicillin was produced on a huge scale and used to treat wounded soldiers and sick patients. It became the first of a group of medicines known as antibiotics.

The discovery of penicillin changed medicine forever. Before antibiotics, people often died from infections that today can be cured easily. Fleming once said that he had not planned to change medicine, but that chance had given him the opportunity. His story is often used to show that great discoveries need not only luck, but also a careful mind ready to notice when something unexpected happens.

(1) What did Alexander Fleming study?

- 1 Very small living creatures that can make people sick.
- 2 The movement of the stars.
- 3 The history of medicine.
- 4 The behavior of large animals.

(2) What did Fleming notice about the dish he had left uncovered?

- 1 The bacteria had grown much larger.
- 2 The dish was completely empty.
- 3 The bacteria near the mold had died.
- 4 The mold had disappeared.

(3) Why was Fleming unable to use penicillin to treat patients at first?

- 1** Because it was too dangerous to use.
 - 2** Because he could not produce it in large enough amounts.
 - 3** Because no one was sick at the time.
 - 4** Because he kept the discovery secret.
-

(4) What did the other scientists do about ten years later?

- 1** They threw away Fleming's notes.
 - 2** They proved that penicillin was useless.
 - 3** They forgot about the discovery.
 - 4** They made penicillin in large amounts and proved it could cure infections.
-

(5) Which of the following is true?

- 1** Fleming planned from the start to change medicine.
- 2** Penicillin was the earliest medicine of the kind now called antibiotics.
- 3** People today often die from infections that have no cure.
- 4** Penicillin was never used to treat soldiers.

3

B

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No.17 / The Surprising History of Chocolate (食・文化史：チョコレートの歴史)

The Surprising History of Chocolate

For most people today, chocolate means something sweet and creamy. But chocolate has a long and surprising history, and for most of that history, it was not sweet at all. Chocolate is made from the beans of the cacao tree, which grows in the hot, rainy parts of Central and South America. Long ago, the people who lived there were the first to discover how to use these beans.

More than a thousand years ago, ancient peoples such as the Maya and, later, the Aztecs roasted and ground cacao beans to make a drink. This drink was nothing like modern hot chocolate. It was bitter and often mixed with spices such as chili pepper, and it was usually drunk without any sugar. Chocolate was so highly valued that the Aztecs even used cacao beans as a kind of money, trading them for goods.

In the sixteenth century, Spanish explorers arrived in the Americas and tasted this strange bitter drink. They carried cacao beans back across the ocean to Europe. At first, Europeans found the drink too bitter for their taste. The big change came when they began adding sugar and honey to it. Sweetened chocolate quickly became a fashionable drink, but because it was expensive, only the rich could afford to enjoy it.

For a long time, chocolate was something people drank, not ate. It was only in the nineteenth century that inventors discovered how to turn it into a solid bar. Later, others found a way to mix in milk, creating the smooth milk chocolate that is popular today. As machines made chocolate cheaper to produce, it slowly changed from a luxury for the wealthy into a treat that almost anyone could buy.

Today, chocolate is enjoyed all over the world in countless forms. Yet many people who eat it have no idea that it began as a bitter drink, thousands of kilometers away and thousands of years ago. The next time you eat a piece of chocolate, it is worth remembering the long journey—across oceans and centuries—that brought it to you.

(1) Where does the cacao tree grow?

- 1 In warm, wet regions of the Americas.
- 2 In the cold, dry mountains of Asia.
- 3 In the deserts of northern Africa.
- 4 Only in southern Europe.

(2) What was the cacao drink made by the Maya and Aztecs like?

- 1 It tasted sweet and creamy, much like chocolate today.
- 2 It had a bitter taste and contained spices but no sugar.
- 3 It was a cold drink made mostly from milk.
- 4 It was a sweet drink flavored only with honey.

-
- (3)** Why did sweetened chocolate become popular in Europe?
- 1** Because it was given away for free.
 - 2** Because it was the only drink available.
 - 3** Because everyone could easily afford it.
 - 4** Because Europeans added sugar, making the bitter drink sweet.
-
- (4)** What change happened to chocolate in the nineteenth century?
- 1** People learned how to make it into a hard, eatable bar.
 - 2** It was banned in many European countries.
 - 3** People stopped drinking it completely.
 - 4** It became far more expensive than before.
-
- (5)** Which of the following is true?
- 1** Chocolate has always been sweet.
 - 2** The Aztecs used gold coins to buy cacao.
 - 3** Chocolate has only ever been a drink.
 - 4** Chocolate began as a bitter drink long ago in the Americas.

3

B

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No.18 / How GPS Helps Us Find Our Way (テクノロジー：GPSの仕組み)

How GPS Helps Us Find Our Way

Not long ago, people who wanted to find their way in an unfamiliar place had to use a paper map or stop and ask for directions. Today, most of us simply look at our phones, which can show exactly where we are and how to get where we want to go. This is possible thanks to a clever system known as GPS, which stands for Global Positioning System.

GPS depends on a network of satellites that circle the Earth high above us. There are about thirty of these satellites, and they are arranged so that, from almost anywhere on the planet, several of them are always overhead. Each satellite constantly sends out radio signals, like tiny invisible messages, that travel down to the Earth at the speed of light.

A GPS device, such as the one inside a smartphone, does not send any signal of its own. Instead, it simply listens for the signals coming from the satellites. By measuring how long each signal takes to arrive, the device can work out how far away each satellite is. If it can receive signals from at least four satellites at once, it can use this information to calculate its exact position on the Earth's surface.

GPS was first built many years ago by the military of the United States, to help its ships, planes, and soldiers find their way. Later, the system was opened up so that anyone in the world could use it, free of charge. Today, GPS does far more than guide drivers. It helps ships and airplanes travel safely, allows farmers to guide their machines across fields, and even helps scientists track the movements of wild animals.

GPS has become such a normal part of life that we rarely stop to think about it. Yet every time a map appears on a phone screen, signals are racing down from satellites far out in space and being turned into a simple blue dot. A system once built for armies now quietly helps billions of ordinary people find their way every single day.

(1) How did people find their way in unfamiliar places not long ago?

- 1 By sending signals to satellites.
- 2 By looking at a printed map or asking someone the way.
- 3 By staying at home.
- 4 By following wild animals.

(2) What do the GPS satellites do?

- 1 They take photographs of people.
- 2 They travel along the ground.
- 3 They listen for signals from phones.
- 4 They keep sending radio messages toward the Earth.

-
- (3)** How does a GPS device work out where it is?
- 1** By measuring how long signals take to arrive from several satellites.
 - 2** By sending a strong signal into space.
 - 3** By asking nearby phones for help.
 - 4** By using a paper map inside it.
-

- (4)** Who first built the GPS system?
- 1** A group of farmers.
 - 2** A team of scientists studying animals.
 - 3** The armed forces of the United States.
 - 4** A company that makes phones.
-

- (5)** Which of the following is true?
- 1** GPS can now be used by people everywhere without paying.
 - 2** A GPS device must send a signal to the satellites.
 - 3** GPS can only be used by drivers.
 - 4** GPS satellites travel along the Earth's surface.

3

B

次の英文の内容に関して、(1) から (5) までの質問に対して最も適切なもの、または文を完成させるのに最も適切なものを 1, 2, 3, 4 の中から一つ選びなさい。

No.19 / The Great Fire of London (歴史的事件：ロンドン大火)

The Great Fire of London

In the early hours of a September morning in 1666, a small fire broke out in a baker's shop in the city of London. Fires were common in those days, and at first no one was very worried. But this fire would not stay small. Over the next few days, it grew into one of the most destructive fires in the history of England, changing the city forever.

Several things helped the fire spread so quickly. Most of the houses in London at that time were built of wood, and they stood very close together along narrow streets. The summer had been long and dry, so the wood was easy to burn. Worst of all, a strong wind was blowing, carrying burning pieces from one building to the next. Once the fire took hold, it was almost impossible to stop.

The fire burned for about four days. By the time it finally died down, it had destroyed a huge part of the city. Thousands of houses were burned, along with many churches and the great cathedral of St Paul's. Tens of thousands of people lost their homes and had to camp in the open fields outside the city. Yet, surprisingly, records suggest that very few people were killed in the fire.

After such a disaster, the people of London had to rebuild their city almost from nothing. This gave them a chance to make important changes. New rules said that buildings should be made of brick or stone instead of wood, and that streets should be wider, so that fire could not spread so easily again. A famous architect was chosen to design a beautiful new St Paul's Cathedral, which still stands today.

The Great Fire also led to other lasting changes. To help people who lost everything, the first fire insurance companies were created, allowing people to pay a small sum so they would be helped if their home burned. Organized teams to fight fires also began to develop. In this way, a terrible disaster led to a safer, better-built city—and to ideas that we still rely on today.

(1) Where did the Great Fire of London begin?

- 1 In a baker's shop.
- 2 In a church.
- 3 In a field outside the city.
- 4 In the king's palace.

(2) What was one reason the fire spread so quickly?

- 1 The houses were made of brick.
- 2 Buildings of wood were packed together along narrow streets.
- 3 There had been heavy rain.
- 4 There was no wind at all.

(3) What does the passage say about the number of people killed?

- 1 Tens of thousands of people died.
- 2 Half the city's people died.
- 3 Only a very small number of people lost their lives.
- 4 No one lost their home.

-
- (4)** What new rules were made when London was rebuilt?
- 1** Streets should be made narrower.
 - 2** Houses should be built of wood.
 - 3** The city should not be rebuilt at all.
 - 4** Buildings should be of brick or stone, and streets wider.
-
- (5)** Which of the following is true?
- 1** The fire destroyed only a few houses.
 - 2** Heavy rain helped put the fire out at once.
 - 3** The disaster led to the first fire insurance companies.
 - 4** The new St Paul's Cathedral has since been torn down.

3

B

次の英文の内容に関して、(1) から (5) までの質問に対して最も適切なもの、または文を完成させるのに最も適切なものを 1, 2, 3, 4 の中から一つ選びなさい。

No.20 / The Helpful Bacteria Inside Us (体のしくみ: 腸内細菌)

The Helpful Bacteria Inside Us

When we hear the word "bacteria," we usually think of germs that make us sick. It may be surprising, then, to learn that our bodies are home to trillions of tiny living things, most of them bacteria—and that the great majority of them are not harmful at all. In fact, many of them are extremely important for keeping us healthy. The largest number of these helpers live inside our intestines, the part of the body that digests our food.

These tiny helpers do several important jobs. One of the most important is helping us digest our food. There are some parts of the food we eat, especially in vegetables and grains, that our bodies cannot break down on their own. The bacteria in our gut are able to break these parts down for us, releasing useful energy and nutrients that we would otherwise lose.

The work of these bacteria does not stop there. Some kinds make vitamins that our bodies need but cannot easily produce by themselves. Just as importantly, the bacteria help to train our immune system, the body's defense against illness. By living alongside these friendly bacteria from an early age, the body learns to tell the difference between harmless visitors and the dangerous germs it needs to fight.

In recent years, scientists have become very interested in these gut bacteria. They have found that the particular mix of bacteria a person has may affect their health in many ways, from how well they digest food to how easily they fight off illness. Some research even suggests that gut bacteria may have an effect on a person's mood, though scientists are still working hard to understand exactly how this happens.

The good news is that we can help our friendly bacteria stay healthy. They feed on fiber, which is found in foods such as vegetables, fruit, beans, and whole grains. Eating a wide variety of these foods gives the helpful bacteria what they need to do their work. In other words, looking after the tiny creatures inside us is one more reason to eat a healthy, varied diet.

(1) What is surprising about the bacteria in our bodies?

- 1 They are all dangerous germs.
- 2 Most of them are not harmful and help keep us healthy.
- 3 There are very few of them.
- 4 They live only on our skin.

(2) How do gut bacteria help with digestion?

- 1 They make our food taste better.
- 2 They stop us from feeling hungry.
- 3 They break down parts of food our bodies cannot.
- 4 They remove all bacteria from our food.

(3) According to the passage, what is one other job the bacteria do?

- 1** They make our hair grow.
 - 2** They help us see in the dark.
 - 3** They cause most illnesses.
 - 4** They help train the body's immune system.
-

(4) What have scientists recently found about gut bacteria?

- 1** The kinds of bacteria someone has can influence their health in several ways.
 - 2** Gut bacteria have no effect on health.
 - 3** Everyone has exactly the same bacteria.
 - 4** Gut bacteria can be seen with the naked eye.
-

(5) Which of the following is true?

- 1** All bacteria in the body cause disease.
- 2** Our bodies can digest all foods without any help.
- 3** Scientists fully understand how gut bacteria affect mood.
- 4** Eating fiber-rich foods helps the friendly bacteria.

3

B

次の英文の内容に関して、(1) から (5) までの質問に対して最も適切なもの、または文を完成させるのに最も適切なものを 1, 2, 3, 4 の中から一つ選びなさい。

No.21 / Why Music Is Good for Us (音楽・脳科学：音楽の力と音楽療法)

Why Music Is Good for Us

Every human culture in the world makes music. From quiet lullabies to loud dance songs, music is a part of life everywhere, and it has been for thousands of years. But music is more than just entertainment. Scientists who study the brain have discovered that listening to and making music can have powerful effects on both our minds and our bodies.

One of the clearest effects of music is on our mood. When we hear a song we love, our brains release chemicals that make us feel pleasure and happiness—the same kind of chemicals released when we enjoy good food or meet a friend. This is why a favorite song can lift our spirits on a difficult day, and why sad music can comfort us when we feel low.

Music is also closely linked to memory. A tune we have not heard for years can suddenly bring back strong memories of a place, a person, or a feeling from long ago. Doctors have found this link especially helpful for people who have lost much of their memory through illness. Such patients may have trouble remembering recent events, yet they can still sing along to songs they loved when they were young.

Because of effects like these, music is now used to help people heal, in a practice called music therapy. Music can lower stress and even reduce the feeling of pain in hospital patients. It can also help in surprising ways: some people who have lost the ability to speak after a brain injury find that they can still sing, and singing can slowly help them learn to speak again.

Music, then, is far more than a pleasant sound. It can change how we feel, unlock our memories, and even help our bodies recover. The next time a song makes you want to dance, or brings a tear to your eye, remember that something remarkable is happening inside your brain. Perhaps this is why, in every corner of the world, human beings have never stopped making music.

(1) What have scientists who study the brain discovered about music?

- 1 That only some cultures make music.
- 2 That music can powerfully affect our minds and bodies.
- 3 That music is only for entertainment.
- 4 That music is bad for the brain.

(2) Why can a favorite song lift our spirits?

- 1 Because it brings back memories of a difficult day.
- 2 Because it makes us want to eat good food.
- 3 Because it produces chemicals in the brain that bring us pleasure.
- 4 Because it teaches the brain how to relax slowly.

(3) What have doctors found about music and people who have lost their memory?

- 1 They can often still sing songs they loved when young.
- 2 They can no longer enjoy any music.
- 3 They forget music faster than anything else.
- 4 They can suddenly remember every recent event.

(4) How can music therapy help some people who have lost the ability to speak?

- 1** By teaching them to read.
 - 2** By helping them sleep more.
 - 3** Singing can help them slowly learn to speak again.
 - 4** By stopping them from singing.
-

(5) Which of the following is true?

- 1** Music can affect our feelings, memory, and even our health.
- 2** Music is found in only a few cultures.
- 3** Music has no effect on stress or pain.
- 4** Sad music always makes people feel worse.

3

B

次の英文の内容に関して、(1) から (5) までの質問に対して最も適切なもの、または文を完成させるのに最も適切なものを 1, 2, 3, 4 の中から一つ選びなさい。

No.22 / How Bats Find Their Way in the Dark (動物の能力：エコーロケーション)

How Bats Find Their Way in the Dark

Bats are among the few animals that are active mostly at night. Flying through the darkness, they must somehow find their food and avoid crashing into trees, walls, and each other. Yet many kinds of bats cannot rely on their eyes to do this, because there is simply too little light. Instead, they use one of the most amazing tricks in the animal world: they "see" with sound.

As a bat flies, it makes a stream of very high-pitched sounds. These sounds are so high that human ears cannot hear them at all. When the sounds hit an object—a tree, a wall, or a tiny insect—they bounce back to the bat as echoes. By listening carefully to these echoes, the bat can tell where the object is, how far away it is, and even how big it is. This clever skill is called echolocation.

Echolocation is astonishingly accurate. Using only sound, a bat flying at full speed in complete darkness can find and catch a flying insect smaller than a fingernail. It can also tell the difference between a real insect and a falling leaf, and steer neatly around obstacles it has never seen. In a single second, a hunting bat may send out and listen to many separate sounds, building a kind of "sound picture" of the world around it.

Bats are not the only animals that use sound in this way. Dolphins, for example, use a similar method to find fish in the dark ocean. Humans, too, have copied the idea. Ships and submarines use a system called sonar, which sends out sounds and listens for the echoes, to find objects underwater and measure the depth of the sea. In a sense, nature discovered this trick long before people did.

It is easy to feel sorry for an animal that cannot see well in the dark. But bats are not helpless at all. With their remarkable use of sound, they are perfectly suited to a life in the night sky, hunting and flying with a skill that, in some ways, is better than sight. The bat reminds us that there is more than one way to understand the world.

(1) Why can many bats not use their eyes to find food at night?

- 1 Because they have lost the ability to see.
- 2 Because the night is too dark for their eyes to work.
- 3 Because they keep their eyes closed while flying.
- 4 Because they hunt only during the daytime.

(2) What is echolocation?

- 1 Using the echoes of high sounds to find objects.
- 2 Using bright light to see in the dark.
- 3 Smelling food from far away.
- 4 Following other bats by sight.

(3) According to the passage, what can a bat do using only sound?

- 1** It can read words written on a wall.
 - 2** It can only find very large objects.
 - 3** It can see colors in the dark.
 - 4** It can catch a tiny flying insect in complete darkness.
-

(4) How do ships and submarines use a method like the bat's?

- 1** They send out light to see underwater.
 - 2** They use sonar, sending sounds and listening for echoes.
 - 3** They follow dolphins to find fish.
 - 4** They make sounds too low for fish to hear.
-

(5) Which of the following is true?

- 1** Bats are helpless in the dark.
- 2** Only bats use sound to find their way.
- 3** Humans invented echolocation before animals used it.
- 4** Dolphins use a method similar to the bat's to find fish.

3

B

次の英文の内容に関して、(1) から (5) までの質問に対して最も適切なもの、または文を完成させるのに最も適切なものを 1, 2, 3, 4 の中から一つ選びなさい。

No.23 / Living with Earthquakes (防災・地学：地震とともに生きる／日本)

Living with Earthquakes

An earthquake can be one of the most frightening events in nature. In just a few seconds, the ground that we trust to be solid begins to shake, sometimes strongly enough to damage buildings and roads. To understand why earthquakes happen, we have to look deep beneath our feet. The Earth's hard outer surface is not a single piece; it is broken into several enormous sections, called plates, which move very slowly all the time.

Most earthquakes happen where two of these plates meet. As the plates push against each other, their rough edges can become stuck, while the rest of the plate keeps trying to move. Over many years, pressure builds up at the stuck point, like a spring being slowly pressed. Finally, the rock can take no more, and the edges suddenly slip past each other. This sudden movement releases a huge amount of energy, which travels through the ground as the shaking we feel.

Because earthquakes happen mainly where plates meet, some parts of the world have far more of them than others. Japan, for example, sits on the edges of several plates, which is why it experiences many earthquakes, both large and small. People who live in such places must learn to live with the knowledge that a strong earthquake could come at any time, and prepare themselves as well as they can.

Despite years of study, scientists still cannot say exactly when or where the next big earthquake will strike. However, there is much that people can do to stay safe. Engineers now design buildings that can bend slightly and sway without falling. Schools and offices hold regular earthquake drills so that people know what to do. Many families keep an emergency kit with water, food, and a flashlight, ready in case help cannot reach them quickly.

One of the most useful tools is the earthquake early-warning system. Special instruments can detect the first, faster waves of an earthquake and send out an alert seconds before the strong shaking arrives. A few seconds may not sound like much, but it can be enough time to take cover, stop a moving train, or turn off dangerous machines. Thanks to such efforts, people are better able to protect themselves than ever before.

(1) What does the passage say about the Earth's hard outer surface?

- 1 It is one solid piece of rock that never moves.
- 2 It is made up of several huge parts that keep moving slowly.
- 3 It is covered by a thick layer of water.
- 4 It is formed from many small loose stones.

(2) Why does pressure build up before an earthquake?

- 1 The plates stop moving against each other completely.
- 2 Heavy rain makes the ground beneath the plates wet.
- 3 The edges of the plates get stuck while the rest keeps trying to move.
- 4 The rock slowly becomes much hotter than before.

(3) What is the reason that Japan has so many earthquakes?

- 1** It is located where several plates come together.
 - 2** It has a large number of very tall buildings.
 - 3** It is one of the smallest countries in the world.
 - 4** It has fewer plates than most other countries.
-

(4) What is one thing people do to stay safer during earthquakes?

- 1** They work out the exact day that the next quake will come.
 - 2** They put up houses that are built not to move at all.
 - 3** They move away from areas where plates meet.
 - 4** They make buildings that can move a little without falling down.
-

(5) What can an earthquake early-warning system do?

- 1** It tells scientists exactly where the next quake will happen.
- 2** It gives people a short warning before the strong shaking begins.
- 3** It stops earthquakes from causing any damage at all.
- 4** It detects only the slower waves that arrive last.

3

B

次の英文の内容に関して、(1) から (5) までの質問に対して最も適切なもの、または文を完成させるのに最も適切なものを 1, 2, 3, 4 の中から一つ選びなさい。

No.24 / The Invention That Spread Ideas (発明・歴史：活版印刷)

The Invention That Spread Ideas

Today, books are everywhere, and most of us can buy one cheaply or borrow it from a library. For most of history, however, this was not the case. Before the middle of the fifteenth century, every single book had to be copied out by hand, usually by people who spent months bent over their desks, writing one page at a time. As a result, books were extremely rare and expensive, and only the wealthy or the highly educated owned them.

This changed because of a German named Johannes Gutenberg. In the 1440s, he developed a machine known as the printing press. Its clever feature was the use of small metal letters that could be moved around. A worker could arrange these letters to form the words of a page, cover them with ink, and press paper onto them to make a printed sheet. The same letters could then be rearranged to print a completely different page.

The effect of this invention was enormous. Instead of taking months to copy one book, a printer could now produce many copies of the same book in a fraction of the time. As more books were printed, their price fell, and ordinary people could finally afford to own them. With more books available, far more people had a reason to learn to read.

The printing press did more than make books cheaper; it changed how ideas traveled. New thoughts, discoveries, and stories could now be printed and carried across countries within weeks, rather than slowly spreading by word of mouth. Knowledge that had once been kept by a small number of people could now reach thousands. Many historians believe that this rapid spread of ideas helped bring about huge changes in science, religion, and society.

Gutenberg himself did not become rich from his invention, and he probably never imagined how important it would become. Yet the printing press is now seen as one of the most powerful inventions in human history. It opened the world of reading to ordinary people and showed that an idea, once printed, could travel further and faster than anyone had thought possible.

(1) Before the fifteenth century, why were books rare and expensive?

- 1 Because making each one required a lot of slow handwork.
- 2 Because very few people were interested in reading them.
- 3 Because the materials needed to make them did not yet exist.
- 4 Because too many copies were being produced at once.

(2) What was the clever feature of Gutenberg's printing press?

- 1 It wrote out the words of each page with a pen.
- 2 It could be used to print only a single page.
- 3 Its metal letters could be set up again in a new order.
- 4 It worked without using any ink or paper.

-
- (3)** Why did more people have a reason to learn to read?
- 1** Reading and owning books became possible for ordinary people.
 - 2** The cost of buying a book kept rising.
 - 3** People were required by law to study reading.
 - 4** Fewer books were produced than in earlier times.
-
- (4)** How did the printing press change the way ideas traveled?
- 1** It prevented new discoveries from becoming known.
 - 2** It limited the sharing of ideas to spoken conversation.
 - 3** It kept useful knowledge in the hands of a few.
 - 4** It allowed information to reach distant places in a short time.
-
- (5)** Which of the following is true?
- 1** Gutenberg earned a large fortune through his invention.
 - 2** The printing press is regarded as a deeply influential invention.
 - 3** Books are still produced by copying them out by hand today.
 - 4** The arrival of the press caused the price of books to rise.

3

B

次の英文の内容に関して、(1) から (5) までの質問に対して最も適切なもの、または文を完成させるのに最も適切なものを 1, 2, 3, 4 の中から一つ選びなさい。

No.25 / The Mystery of Black Holes (宇宙の謎：ブラックホール)

The Mystery of Black Holes

Among the many wonders of space, few are as strange or as fascinating as the black hole. A black hole is a place where the pull of gravity is so incredibly strong that nothing nearby can escape from it—not even light. Because light cannot get out, a black hole gives off no light of its own, which is exactly why it appears completely black and why it is so difficult to study.

Where do black holes come from? Many of them are born when a very large star reaches the end of its life. While a star is alive, it burns fuel that pushes outward and keeps it from collapsing. When a giant star finally runs out of fuel, there is nothing left to hold it up. Its huge weight causes it to fall inward on itself, squeezing an enormous amount of material into a tiny space and creating a region with incredibly powerful gravity.

If a black hole cannot be seen, how do scientists know it is there? The answer is that they watch its effects on the things around it. As stars and gas are pulled toward a black hole, they move in unusual ways and grow extremely hot, giving off other kinds of energy that our instruments can detect. By studying how nearby objects behave, scientists can work out that a black hole must be hiding in that spot, even though they cannot see it directly.

Black holes are not as rare as they may sound. Scientists now believe that a giant black hole sits at the center of our own galaxy, the Milky Way, as well as at the heart of many other galaxies. For a long time, no one had ever seen anything close to a black hole. Then, in recent years, scientists joined together telescopes from around the world and managed to capture the first-ever image of the glowing ring of hot material surrounding one.

Even with such progress, black holes remain one of the great mysteries of the universe. Scientists are still trying to understand exactly what happens to anything that falls inside one, and what these strange objects can teach us about space, time, and gravity. The more we learn about black holes, the more questions they seem to raise.

(1) Why does a black hole appear completely black?

- 1 Because it is very far away.
- 2 Because it is covered in dark dust.
- 3 Because it is very cold.
- 4 Because no light can come out of it.

(2) How are many black holes formed?

- 1 When a very large star runs out of fuel and collapses.
- 2 When two small stars join together gently.
- 3 When a planet grows too large.
- 4 When light is trapped by a cloud.

(3) How do scientists know a black hole is there if they cannot see it?

- 1** By traveling to it in a spaceship.
 - 2** By watching how nearby stars and gas behave.
 - 3** By listening for sounds from space.
 - 4** By measuring its color.
-

(4) What did scientists recently manage to do?

- 1** They flew inside a black hole.
 - 2** They removed a black hole from the sky.
 - 3** They captured the first image of the glowing material around a black hole.
 - 4** They proved black holes do not exist.
-

(5) Which of the following is true?

- 1** Light can easily escape from a black hole.
- 2** Black holes are found only outside our galaxy.
- 3** Our own galaxy is thought to have a huge black hole at its center.
- 4** Everything about black holes is now fully understood.

3

B

次の英文の内容に関して、(1) から (5) までの質問に対して最も適切なもの、または文を完成させるのに最も適切なものを 1, 2, 3, 4 の中から一つ選びなさい。

No.26 / Exploring the Deep Sea (海・科学：深海の探検)

Exploring the Deep Sea

The ocean covers most of our planet, yet much of it remains a complete mystery. While we have made detailed maps of the Moon and Mars, we have explored only a tiny part of the deep sea. The deepest parts of the ocean lie several kilometers below the surface, in a world that is almost completely unknown to us. In many ways, the bottom of the ocean is one of the last unexplored places on Earth.

The deep sea is so hard to explore because it is one of the harshest places imaginable. Below a certain depth, no sunlight can reach, so the deep ocean is wrapped in total darkness. It is also extremely cold. Worst of all is the pressure: the huge weight of all the water above pushes down so hard that it would instantly crush most ordinary machines. To go there, scientists need special vehicles built from very strong materials.

For a long time, people thought that nothing could live in such a place. They were wrong. The deep sea is home to many strange creatures, and one of their most amazing features is that many of them can make their own light. This glowing, called bioluminescence, helps them in different ways: some use it to attract food, others to find a mate, and some to confuse the animals that hunt them, all in a place where there is no other light at all.

Even more surprising are the creatures that live near cracks in the ocean floor, where hot water full of minerals pours out from inside the Earth. Almost all life on the surface depends on energy from the sun. But around these hot vents, deep in the darkness, special living things get their energy from the chemicals in the water instead. The discovery of these creatures showed scientists that life can survive in ways they had never imagined.

Exploring the deep sea is difficult, slow, and expensive, but scientists believe it is worth the effort. Each journey to the ocean floor brings the discovery of new creatures and teaches us more about how life can exist in extreme places. The deep sea reminds us that, even on our own planet, there are still vast worlds waiting to be explored.

(1) What does the passage say about how much of the deep sea we have explored?

- 1 We have learned about almost none of it so far.
- 2 We have studied it more closely than the Moon.
- 3 We have finished mapping all of its deepest parts.
- 4 We understand it as well as we understand Mars.

(2) Why would the deep sea instantly crush most ordinary machines?

- 1 The water there moves with very strong currents.
- 2 The freezing temperatures make metal break apart.
- 3 The water above presses down on them with enormous force.
- 4 The lack of light damages their cameras and lights.

(3) What is one way deep-sea creatures use the light they make?

- 1** They use it to keep their bodies warmer.
 - 2** They use it to draw prey toward them.
 - 3** They use it to make the dark water clearer.
 - 4** They use it to swim through the water faster.
-

(4) How do the creatures near the hot vents get their energy?

- 1** From weak sunlight that reaches the ocean floor.
 - 2** By eating dead animals that sink from the surface.
 - 3** From substances dissolved in the water around them.
 - 4** By producing light from inside their own bodies.
-

(5) Which of the following is true about the deep sea?

- 1** No living things are able to survive down there.
- 2** Bright sunlight reaches even its deepest parts.
- 3** Scientists know it better than they know the Moon.
- 4** A number of its creatures are able to produce light.

3

B

次の英文の内容に関して、(1) から (5) までの質問に対して最も適切なもの、または文を完成させるのに最も適切なものを 1, 2, 3, 4 の中から一つ選びなさい。

No.27 / The Art of Origami (日本・芸術×科学：折り紙)

From Paper Cranes to Space: The Art of Origami

Origami is the traditional Japanese art of folding paper into shapes and figures. The word itself comes from Japanese, joining the words for "fold" and "paper." In its purest form, origami uses a single square sheet of paper, which is folded—without any cutting or gluing—into beautiful and often complicated designs. The most famous of all origami models is probably the paper crane, a graceful bird that many people learn to fold as children.

Origami has been enjoyed in Japan for hundreds of years. Long ago, because paper was expensive, folded-paper designs were often used only on special occasions, such as ceremonies and gifts. Over time, as paper became cheaper and more common, origami grew into a popular pastime and art form for everyone. Today, people all over the world fold paper for fun, and skilled artists create amazing works, from tiny insects to huge, lifelike animals.

In recent years, origami has caught the attention of an unexpected group of people: scientists and engineers. At first, this may seem strange—what could paper folding have to do with science? The answer is that origami solves a very useful problem. It shows how something large and flat can be folded into a small, neat shape, and then opened out again to its full size when needed.

This simple idea has led to some remarkable inventions. For example, engineers have used origami methods to fold up large solar panels so that they fit inside a rocket, then unfold once they are out in space. Doctors, too, have designed tiny medical devices that can be inserted into the body while folded, and then expanded to do their work. In these ways, an ancient art is helping to solve very modern problems.

Origami is a wonderful example of how art and science can meet. What began as a way of folding paper for beauty and pleasure has turned out to hold lessons useful in space travel and medicine. The next time you fold a paper crane, you are taking part in a tradition that is centuries old—and, perhaps, touching ideas that are shaping the future.

(1) What does origami in its purest form NOT involve?

- 1 Using a square piece of paper.
- 2 Cutting or gluing the paper.
- 3 Folding the paper into designs.
- 4 Making shapes such as birds.

(2) Why was folded paper used only on special occasions long ago?

- 1 Folding paper was against the law.
- 2 Paper cost a great deal of money.
- 3 Few people knew how to fold it.
- 4 It was meant only for children.

-
- (3)** Why have scientists and engineers become interested in origami?
- 1** It can be done only by people in Japan.
 - 2** It makes use of very expensive paper.
 - 3** It can turn a big, flat object into a small one that can later be opened up again.
 - 4** It is one of the hardest skills to master.
-
- (4)** What is one way engineers have used origami methods?
- 1** To lower the price of paper.
 - 2** To pack solar panels tightly so they can be carried into space.
 - 3** To create new types of books.
 - 4** To help children learn to read.
-
- (5)** What is one thing the writer wants to tell readers?
- 1** Origami can never be done without cutting the paper.
 - 2** Origami is something people enjoy only in Japan.
 - 3** Origami is a brand-new art that started just a short time ago.
 - 4** An old art such as origami can help solve modern problems.

3

B

次の英文の内容に関して、(1) から (5) までの質問に対して最も適切なもの、または文を完成させるのに最も適切なものを 1, 2, 3, 4 の中から一つ選びなさい。

No.28 / The World's Oldest Art (芸術・歴史：洞窟壁画)

The World's Oldest Art

Long before people could read or write, and long before the first cities were built, human beings were already making art. Tens of thousands of years ago, our distant ancestors painted pictures on the walls of caves. Some of the most famous of these paintings have been found in caves in France and Spain, but ancient paintings exist on many other continents too. They are among the oldest works of art in the world.

What did these early artists paint? Most often, they painted the animals around them. On the cave walls we can see horses, deer, and large wild cattle, sometimes shown running or being hunted. The paintings are surprisingly skillful, capturing the shape and movement of the animals with just a few lines. Some caves contain dozens of these images, covering the walls and even the ceilings.

Making these paintings was not easy. The early artists had no shops to buy paint from, so they made their own colors from natural materials, such as colored earth, ground-up rock, and charcoal from their fires. They applied the paint with their fingers, with simple brushes, or by blowing colored powder through hollow bones. Because many of the paintings are deep inside dark caves, the artists must have worked by the weak light of simple lamps.

One of the biggest questions is why people made these paintings at all. No one knows the answer for certain. Some experts think the pictures were part of religious or magical ceremonies—perhaps the painters hoped that drawing an animal would bring them luck in the hunt. Others believe the images were a way of telling stories or sharing knowledge. It is also possible that early people painted simply because, like us, they enjoyed creating beautiful things.

Whatever their purpose, cave paintings are precious to us today. They give us a rare window into the lives of people who lived in a world without writing, machines, or modern tools. Above all, they show that the desire to create art is extremely old, and may be one of the things that has always made us human.

(1) According to the first paragraph, when did human beings begin making art?

- 1 Soon after the earliest cities appeared.
- 2 Once people had learned to read and write.
- 3 Before writing or cities existed.
- 4 Only after modern tools were invented.

(2) What were the early artists most likely to draw?

- 1 The creatures that lived near them.
- 2 Maps showing the way to their homes.
- 3 Patterns of the stars in the night sky.
- 4 Simple signs that stood for words.

-
- (3)** How did the early artists get the colors they needed?
- 1** They traded for them with other groups.
 - 2** They prepared them from things found in nature.
 - 3** They mixed nothing but plain water.
 - 4** They managed without using any color.
-
- (4)** What does the passage suggest about the reason for the paintings?
- 1** Scientists have now proved why they were made.
 - 2** The true purpose remains uncertain today.
 - 3** They were certainly drawn to guide hunters.
 - 4** They were created to help children learn to read.
-
- (5)** Which of the following is true about cave paintings?
- 1** They were produced after towns and cities had grown.
 - 2** They offer a rare glimpse of life long ago.
 - 3** They prove that early people could already write.
 - 4** Their meaning is now completely understood.

3

B

次の英文の内容に関して、(1) から (5) までの質問に対して最も適切なもの、または文を完成させるのに最も適切なものを 1, 2, 3, 4 の中から一つ選びなさい。

No.29 / Why Do We Dream? (脳・体の謎：なぜ夢を見るのか)

Why Do We Dream?

Almost everyone has woken up from a dream, sometimes a happy one, sometimes a frightening one. Some people say they never dream at all, but scientists believe this is not true. In fact, nearly all of us dream several times every night. The reason some people think they do not dream is simply that they forget their dreams very quickly after waking up.

Sleep is not the same all night long; it passes through different stages. Most of our most vivid dreams happen during a stage called REM sleep, which stands for "rapid eye movement." During this stage, our eyes move quickly behind our closed eyelids, and our brains become almost as active as when we are awake. It is during these active periods that the most colorful and detailed dreams take place.

The biggest question of all is why we dream. Despite many years of research, scientists are still not completely sure of the answer. Dreams are difficult to study, because we cannot see another person's dreams, and because people often forget them. As a result, there are several different ideas about what dreams are for, and no single idea has been proven beyond doubt.

One popular idea is that dreaming helps the brain sort through and store the memories of the day, keeping what is useful and clearing away what is not. Another suggestion is that dreams allow us to practice dealing with difficult or dangerous situations safely, while we sleep. A third idea is that dreams are simply the brain trying to make sense of random signals during sleep. It is even possible that dreaming does several of these things at once.

Whatever the true reason, dreaming appears to be a normal and healthy part of sleep. Far from being a waste of time, our nightly dreams may be quietly helping our brains to work better. The world of dreams remains mysterious, but it is a reminder that even while we rest, our minds are busy and full of activity.

(1) Why do some people think they never dream?

- 1 They do not get enough hours of sleep.
- 2 They lose the memory of their dreams soon after they wake up.
- 3 Their brains really stop dreaming completely.
- 4 They only dream a single time each year.

(2) When do most of our vivid dreams happen?

- 1 Right before we fall asleep.
- 2 Only during the early part of the evening.
- 3 In the period of sleep when the eyes move rapidly.
- 4 At times when we are completely awake.

(3) Why are dreams difficult to study?

- 1 They cannot be watched from outside, and people soon forget them.
- 2 They take place mainly in the daytime.
- 3 Nobody actually has any dreams.
- 4 Scientists have no interest in them at all.

(4) What is one idea about why we dream?

- 1** Dreaming stops the brain from working.
 - 2** Dreaming makes us forget everything.
 - 3** Dreaming keeps us awake at night.
 - 4** Dreaming may help the brain keep information worth saving.
-

(5) Which of the following is true?

- 1** Researchers have not yet found a certain reason for dreaming.
- 2** Dreams have been fully explained by scientists.
- 3** Only a few people dream at night.
- 4** Dreaming is a waste of time for the brain.

3

B

次の英文の内容に関して、(1) から (5) までの質問に対して最も適切なもの、または文を完成させるのに最も適切なものを 1, 2, 3, 4 の中から一つ選びなさい。

No.30 / The Story of the Olympic Games (歴史・文化：オリンピックの物語)

The Story of the Olympic Games

The Olympic Games are the largest sporting event in the world, but their story began more than two thousand years ago, in ancient Greece. In those days, athletes from different Greek cities gathered to compete in contests such as running and wrestling. The games were held in honor of the Greek gods, and winning was considered a very great honor, bringing fame to both the athlete and his home city.

One of the most remarkable things about the ancient games was their connection to peace. At that time, the Greek cities often fought one another. Yet during the period of the Olympic Games, these wars were sometimes paused, so that athletes and spectators could travel safely to the games and back home again. For a short time, the spirit of competition on the sports field replaced fighting on the battlefield.

After many centuries, the ancient Olympic Games came to an end and were not held for a very long time. Then, a little over a hundred years ago, a Frenchman named Pierre de Coubertin had a powerful idea: to bring back the Olympics as a modern, international event. He hoped that, by competing peacefully in sport, people from different nations might learn to understand one another better. The first modern Olympic Games were held in 1896.

Since then, the Olympics have grown enormously. Today they are held every few years, each time in a different host city around the world. Thousands of athletes from almost every country take part, competing in a huge range of sports. Millions more watch on television and online. For many athletes, simply taking part in the Olympics—not only winning a medal—is the achievement of a lifetime.

Although the modern Olympics are very different from the ancient ones, they still carry a similar message. The Games are meant to be a symbol of friendship and peaceful competition between nations. For a few weeks, people from all over the world come together, not to fight, but to celebrate sport, effort, and the things that human beings everywhere have in common.

(1) Why were the ancient Olympic Games held?

- 1 To raise money for Greek cities.
- 2 To choose the leaders of Greece.
- 3 To train soldiers for war.
- 4 To show respect to the gods of Greece.

(2) What was remarkable about the ancient games and peace?

- 1 Wars were started during the games.
- 2 Fighting between cities was sometimes stopped while the games went on.
- 3 The games were held to celebrate winning a war.
- 4 No one was allowed to travel during the games.

(3) What was Pierre de Coubertin's idea?

- 1** To stop all sporting events.
 - 2** To keep the Olympics only in Greece.
 - 3** To revive the ancient games as a worldwide modern event.
 - 4** To hold the games every year in one city.
-

(4) What does the passage say about taking part in the modern Olympics?

- 1** For many athletes, just competing is the greatest accomplishment of their life.
 - 2** Only winning a medal matters to athletes.
 - 3** Very few athletes take part each time.
 - 4** The games are held in the same city every time.
-

(5) Which of the following is true?

- 1** The Olympic Games began only a hundred years ago.
- 2** The ancient games had nothing to do with peace.
- 3** The modern Olympics are always held in Greece.
- 4** The modern Games are intended to stand for friendship and peace among countries.

解答一覧 | Answer Key

No.	トピック	(1)	(2)	(3)	(4)	(5)
1	Florence Nightingale（伝記：近代看護の母）	2	4	1	3	1
2	The Return of the Wolves（科学：イエローストーンの狼）	3	1	4	2	4
3	The Problem of Food Waste（社会問題：食品ロス）	1	3	2	4	2
4	Louis Braille（伝記：点字の発明）	4	2	3	1	3
5	Why Cities Are Hotter（科学・社会：ヒートアイランド）	1	4	2	3	4
6	Wangari Maathai（伝記：植林とノーベル平和賞）	3	1	4	2	1
7	The Secret Life of Octopuses（科学：タコの知能）	2	4	1	3	4
8	健康・社会：10代の睡眠	4	2	3	1	2
9	The Wright Brothers（伝記：動力飛行の発明）	1	3	2	4	3
10	The Trouble with Ocean Plastic（社会・環境：海洋プラスチック）	2	4	1	3	4
11	Living on the Space Station（科学・宇宙：ISSの暮らし）	1	3	2	4	1
12	Hokusai（伝記：葛飾北斎・生涯学び続けた絵師）	3	1	2	3	1
13	The Monarch Butterfly（科学・自然：オオカバマダラの移動）	2	1	3	1	3
14	Eliud Kipchoge（伝記・スポーツ：2時間切りとマインドセット）	2	4	1	2	3
15	How the Bicycle Changed the World（社会史：自転車が社会を変えた）	1	2	3	4	2
16	The Discovery of Penicillin（医学・歴史：ペニシリンの偶然の発見）	1	3	2	4	2
17	The Surprising History of Chocolate（食・文化史：チョコレートの歴史）	1	2	4	1	4
18	How GPS Helps Us Find Our Way（テクノロジー：GPSの仕組み）	2	4	1	3	1
19	The Great Fire of London（歴史的事件：ロンドン大火）	1	2	3	4	3
20	The Helpful Bacteria Inside Us（体のしくみ：腸内細菌）	2	3	4	1	4
21	Why Music Is Good for Us（音楽・脳科学：音楽の力と音楽療法）	2	3	1	3	1
22	How Bats Find Their Way in the Dark（動物の能力：エコーロケーション）	2	1	4	2	4
23	Living with Earthquakes（防災・地学：地震とともに生きる／日本）	2	3	1	4	2
24	The Invention That Spread Ideas（発明・歴史：活版印刷）	1	3	1	4	2
25	The Mystery of Black Holes（宇宙の謎：ブラックホール）	4	1	2	3	3
26	Exploring the Deep Sea（海・科学：深海の探検）	1	3	2	3	4
27	The Art of Origami（日本・芸術×科学：折り紙）	2	2	3	2	4
28	The World's Oldest Art（芸術・歴史：洞窟壁画）	3	1	2	2	2
29	Why Do We Dream?（脳・体の謎：なぜ夢を見るのか）	2	3	1	4	1
30	The Story of the Olympic Games（歴史・文化：オリンピックの物語）	4	2	3	1	4